
UNIT 4 ESTIMATION OF TWO-VARIABLE REGRESSION MODELS*

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4.0 OBJECTIVES

After reading this unit, you will be able to:

- specify the general form of a bivariate regression model;
- outline the principle behind the OLS method of estimation;
- enumerate the assumptions of OLS method;
- explain the OLS estimation method;
- list the properties of OLS estimators with theoretical details;
- illustrate the interpretation of OLS estimates;
- state the expressions for the standard error of OLS method;
- comment on the characteristics of OLS estimators;

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- prove that the OLS estimators are the best linear unbiased estimators (Gauss Markov theorem);
- outline the significance of the coefficient of determination (r^2); and
- describe the procedure of testing hypothesis with an illustration.

4.1 INTRODUCTION

You may be aware that sample survey has many advantages (lower cost, lesser time, limited resources) over a population census. Therefore, we study a sample instead of the universe. We find out the sample characteristics and generalise the sample results to the population. This process is called estimation of parameters. Thus, estimation is the process of recovering the unknown parameters that were used to generate the sample data. You should have a thorough knowledge of inferential statistics, particularly estimation and hypothesis testing given in Unit 30 and Unit 31 of the course MEC 203: Quantitative Methods.

The relationship between variables in a population is given by the population regression function (PRF). The same relationship in a sample is given by the sample regression function (SRF). In simple linear regression, we estimate the population regression function (PRF) from the sample regression function (SRF). There are three methods of estimation: (i) least squares method, (ii) maximum likelihood method, and (iii) method of moments.

4.2 ESTIMATION OF BIVARIATE MODELS

A simple bivariate linear regression model is of the form:

$$Y_i = \beta_1 + \beta_2 X_i + u_i \quad \dots (4.1)$$

Here, Y_i is the dependent variable, X_i is the independent variable and u_i is the error term representing all factors other than X_i affecting Y_i . Here, β_1 and β_2 are the intercept and slope parameters respectively. For instance, Y_i may be the annual savings of the family and X_i their annual income. Let us say that the data is collected for 20 households (i.e., $n = 20$) and is used for estimation of the intercept and the slope coefficient of savings on income. Suppose we get the result as: Savings = 34560 + 0.58 Income. This tells us that each increase of Rs. 1000 in the annual income will increase the annual savings by Rs. 580.

Equation (4.1) represents the PRF which is not directly observable. We therefore estimate it from the SRF.

The estimated regression line is

$$\hat{Y}_i = \hat{\beta}_1 + \hat{\beta}_2 X_i \quad \dots (4.2)$$

In equation (4.2), note that $\hat{Y}_i$ is the expected value of the dependent variable when the independent variable takes the value X_i .

By comparing (4.1) and (4.2), we find that

$$Y_i = \hat{\beta}_1 + \hat{\beta}_2 X_i + \hat{u}_i = \hat{Y}_i + \hat{u}_i$$

$$\therefore \hat{u}_i = Y_i - \hat{Y}_i \quad \dots (4.3)$$

Remember that $\hat{u}_i$ is the residual between the actual and the estimated value of Y . The value of the error term for a particular combination (Y_i, X_i) is not known to us. We get an estimate of the error value, that is called the 'residual'. Thus, there is a difference between the error term (u_i) and the residual ($\hat{u}_i$).

4.2.1 Classical Assumptions

For carrying out the estimation of the parameters, we make the following four assumptions about the stochastic error term.

$$(i) \quad E(u_i) = 0 \quad \dots (4.4)$$

While the error is positive for some observations, it is negative for other observations. It implies that on average, the error is zero. In other words, the sum of errors across the population is zero.

$$(ii) \quad E(u_i^2) = \sigma^2 \quad \dots (4.5)$$

The variance of the error variable is σ^2 . It is constant across the population. For example, the variance of household consumption expenditure is the same for rich households as well as poor households.

$$(iii) \quad E(u_i u_j) = 0 \text{ for } i \neq j \quad \dots (4.6)$$

There is no correlation between the error terms for different observations in a dataset. In other words, the error terms of two observations are independent.

$$(iv) \quad E(X_i u_i) = 0 \quad \dots (4.7)$$

The explanatory variable is not correlated with the error term. This assumption is required for fulfilment of the properties of the estimators ($\hat{\beta}_0, \hat{\beta}_1$) on unbiasedness and consistency. In order to fulfil this condition, we assume that the explanatory variable (regressor) is exogenously given. We often keep the values of X_i fixed in repeated samples.

$$(v) \quad u_i \sim N(0, \sigma^2) \quad \dots (4.8)$$

The error variable u_i follows normal distribution with zero mean and σ standard deviation. For a given value of X , there is a whole range of values of u and a particular value is selected randomly.

The above assumptions are called the classical assumptions. These assumptions indicate the ideal conditions regarding the error variable. In empirical studies, some of the above assumptions may not be fulfilled. As we will see the fifth assumption is not required for the OLS estimation, although it is required for

hypothesis testing. In the case of maximum likelihood estimation, it is necessary to state the probability distribution function of the variables under consideration.

4.2.2 Ordinary Least Squares (OLS) Method

Our objective is to minimise the value of $\hat{u}_i$. For this purpose, we choose those values of $\hat{\beta}_1$ and $\hat{\beta}_2$, for which $\sum \hat{u}_i^2$ is the minimum. We use the term ‘least squares’ because it involves ‘the smallest sum of the squares of the errors’. From Block 3 of the course MEC 203: Quantitative Methods we know the conditions for minimization of a function: (i) first derivative should be equal to zero, and (ii) second derivative should be positive.

Let us take the square of $\hat{u}_i$ and sum it over the sample. From equation (4.3), we obtain

$$\sum \hat{u}_i^2 = \sum (Y_i - \hat{Y}_i)^2 = \sum (Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i)^2$$

As mentioned earlier, we choose the optimal value of $\hat{\beta}_1$ and $\hat{\beta}_2$. Thus, we should take the partial derivatives of $\sum \hat{u}_i^2$ with respect to $\hat{\beta}_1$ and $\hat{\beta}_2$. The term $\sum \hat{u}_i^2$ is called the Residual Sum of Squares (RSS).

The method described below is called ‘ordinary’ least squares (OLS) as it is the simplest of all – it fulfils the classical assumptions regarding the stochastic error term. In Block 3 we will introduce complexities to the OLS method (for example, weighted least squares, two-stage least squares, etc.).

By taking partial derivative of $\sum \hat{u}_i^2$ with respect to $\hat{\beta}_1$, we obtain

$$\begin{aligned} \frac{\partial \sum \hat{u}_i^2}{\partial \hat{\beta}_1} &= \frac{\partial \sum (Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i)^2}{\partial \hat{\beta}_1} \\ &= 2 \cdot (-1) \cdot \sum (Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i) \end{aligned} \quad \dots (4.9)$$

By equating (4.9) to zero, and re-arranging the terms, we obtain

$$\sum (Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i) = 0$$

or,

$$\sum Y_i = n\hat{\beta}_1 + \hat{\beta}_2 \sum X_i \quad \dots (4.10)$$

By taking partial derivative of $\sum \hat{u}_i^2$ with respect to $\hat{\beta}_2$, we obtain

$$\begin{aligned} \frac{\partial \sum \hat{u}_i^2}{\partial \hat{\beta}_2} &= \frac{\partial \sum (Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i)^2}{\partial \hat{\beta}_2} \\ &= 2 \cdot (-1) \cdot \sum (Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i) \cdot X_i \\ &= 2 \cdot (-1) \cdot \sum (X_i Y_i - \hat{\beta}_1 \cdot X_i - \hat{\beta}_2 X_i^2) \end{aligned} \quad \dots (4.11)$$

By equating (4.11) to zero, and re-arranging the terms, we obtain

$$\sum X_i Y_i = \hat{\beta}_1 \sum X_i + \hat{\beta}_2 \sum X_i^2 \quad \dots (4.12)$$

These two equations (4.10) and (4.12) are called the *normal equations* of least squares method. These are two simultaneous linear equations in two unknowns. These can be solved to obtain the values of $\hat{\beta}_1$ and $\hat{\beta}_2$.

By dividing equation (4.10) by the sample size, n , we obtain

$$\bar{Y} = \widehat{\beta}_1 + \widehat{\beta}_2 \bar{X}$$

Therefore,

$$\widehat{\beta}_1 = \bar{Y} - \widehat{\beta}_2 \bar{X} \quad \dots (4.13)$$

If we know the value of $\widehat{\beta}_2$, we can obtain the value of $\widehat{\beta}_1$ from (4.13). After rearrangement of terms in equations (4.10) and (4.11), we obtain

$$\widehat{\beta}_2 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

If we take $x_i = (X_i - \bar{X})$ and $y_i = (Y_i - \bar{Y})$, then

$$\widehat{\beta}_2 = \frac{\sum x_i y_i}{\sum x_i^2} \quad \dots (4.14)$$

4.2.3 Method of Moments

In the method of moments, we equate the sample moments with the population moments. The n^{th} moment of a random variable is the expected value of the n^{th} power. Thus, the first moment of a random variable is its arithmetic mean.

In Sub-Section 4.2.1, while mentioning the classical assumptions of the error term, we consider two population moments. Look at assumptions (4.3) and (4.6). These assumptions are

$$E(u_i) = 0 \quad \dots (4.3)$$

and,

$$E(X_i u_i) = 0 \quad \dots (4.6)$$

Using (4.1), equations (4.3) and (4.6) can be written as:

$$E(Y - \beta_1 - \beta_2 X) = 0 \quad \dots (4.15)$$

and

$$E[X(Y - \beta_1 - \beta_2 X)] = 0 \quad \dots (4.16)$$

Equations (4.15) and (4.16) can be used to obtain the estimators of β_1 and β_2 . Summing up over all the observations of the sample, and dividing by the sample size n , equations (4.15) and (4.16) become:

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \widehat{\beta}_1 - \widehat{\beta}_2 X_i) = 0 \quad \dots (4.17)$$

$$\frac{1}{n} \sum_{i=1}^n X_i (Y_i - \widehat{\beta}_1 - \widehat{\beta}_2 X_i) = 0 \quad \dots (4.18)$$

By re-arranging terms, we can write equation (4.17) as:

$$\bar{Y} = \widehat{\beta}_1 + \widehat{\beta}_2 \bar{X} \quad \dots (4.19)$$

where $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$.

Note that $\bar{Y}$ and $\bar{X}$ in (4.19) are the sample means of Y and X respectively. We therefore have

$$\widehat{\beta}_1 = \bar{Y} - \widehat{\beta}_2 \bar{X} \quad \dots (4.20)$$

Thus, if we have the slope estimator $\widehat{\beta}_2$, we can obtain the intercept estimator $\widehat{\beta}_1$ from equation (4.20) by using the sample values of $\bar{Y}$ and $\bar{X}$. The estimator of the slope coefficient $\widehat{\beta}_2$ can be obtained as follows:

Let us re-write (4.18) [by dropping $\frac{1}{n}$ and substituting for $\widehat{\beta}_1$ from (4.15)] as:

$$\sum_{i=1}^n X_i(Y_i - (\bar{Y} - \widehat{\beta}_2\bar{X}) - \widehat{\beta}_2 X_i) = 0 \quad \dots (4.21)$$

Rearranging (4.21) we get:

$$\sum_{i=1}^n X_i(Y_i - \bar{Y}) = \widehat{\beta}_2 \sum_{i=1}^n X_i(X_i - \bar{X}) \quad \dots (4.22)$$

Note that by the properties of summation, we have:

$$\sum_{i=1}^n X_i(X_i - \bar{X}) = \sum_{i=1}^n (X_i - \bar{X})^2$$

and

$$\sum_{i=1}^n X_i(Y_i - \bar{Y}) = \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

Hence, by assuming that

$$\sum_{i=1}^n (X_i - \bar{X})^2 > 0 \quad \dots (4.23)$$

An estimator of slope coefficient can be obtained as

$$\widehat{\beta}_2 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} \quad \dots (4.24)$$

Or,

$$\widehat{\beta}_2 = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2} \quad \dots (4.25)$$

The estimators of $\widehat{\beta}_1$ and $\widehat{\beta}_2$ in (4.20) and (4.25) respectively are the same as the Ordinary Least Squares (OLS) estimators given at (4.13) and (4.14). From (4.25) we find that the slope estimator $\widehat{\beta}_2$ is simply the sample covariance between X and Y divided by the sample variance of X . So, the only condition required for OLS estimators to exist is that $\sum_{i=1}^n (X_i - \bar{X})^2 > 0$. This condition is always fulfilled, except for the situation where the value of all X_i is the same. Such cases are not common. *Thus, to obtain estimates of the parameters, we should have different values of X .* It is an essential condition in econometric estimation of regression models that $n > k$.

4.2.4 Maximum Likelihood Method

You are familiar with the concept of probability density function such as normal distribution. If we have two random variables (say, X_1 and X_2), then we have a joint probability density function. The likelihood function is identical in functional form with the joint density function. The joint density function is ‘a function of the random variable outcomes’, given the value of the parameters. The interpretation of the likelihood function is the opposite of the joint density function. The likelihood function is ‘a function of the values of the parameters’, given the values of the random variable outcomes.

$$f(X|\theta) = L(\theta|X) \quad \dots (4.26)$$

where $L(\theta|X)$ is the likelihood function.

Let me explain the concept of likelihood function in terms of the error variable u_i . In Sub-Section 4.2.1, we mentioned the assumptions of the error variable u_i . As per assumption (4.8), the stochastic error variable u_i follows normal distribution with mean zero and standard deviation σ such that $u_i \sim N(0, \sigma^2)$.

Suppose we have to select the first observation of the sample. The error term u_1 could assume any random value and its probability density function is given by

$$f(u_1|\theta) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{u_1 - E(u_1)}{\sigma}\right]^2}$$

Since $E(u_1) = 0$, we have

$$f(u_1) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}u_1^2} \quad \dots (4.27)$$

Similarly, the probability density function of the second error term u_2 is given by

$$f(u_2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{u_2 - E(u_2)}{\sigma}\right]^2} = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}u_2^2}$$

As per assumption (iii), the error terms are independent of one another. We can visualize the occurrence of the stochastic error terms (u_i, u_j) as independent events. Therefore, the joint density function of u_1, u_2 is the multiplication of $P(u_1)$ and $P(u_2)$ given above. If we have data on n sample observations, we have $u_1, u_2, u_3, \dots, u_n$. The likelihood of these independent random variables (that is, u_i) is given by

$$L(\theta|u) = \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{u_i - E(u_i)}{\sigma}\right]^2} \quad \dots (4.28)$$

From (4.3) we know that $\hat{u}_i = (Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i)$.

Therefore,

$$L(\theta|u) = \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{Y_i - \beta_1 - \beta_2 X_i}{\sigma}\right]^2} \quad \dots (4.29)$$

If we take natural log of (4.28) we obtain

$$\ln L(\cdot) = -\frac{n}{2} \ln \sigma^2 - \frac{n}{2} \ln(2\pi) - \frac{1}{2} \sum \frac{(Y_i - \beta_1 - \beta_2 X_i)^2}{\sigma^2} \quad \dots (4.30)$$

There are three parameters, viz., β_1, β_2 and σ in (4.30), for which we must obtain estimators. For maximization of the likelihood function, L , we take partial derivatives of (4.32) with respect to $\tilde{\beta}_1, \tilde{\beta}_2$ and $\tilde{\sigma}$.

$$\frac{\partial \ln L}{\partial \tilde{\beta}_1} = \frac{1}{\tilde{\sigma}^2} \sum (Y_i - \tilde{\beta}_1 - \tilde{\beta}_2 X_i)(-1) \quad \dots (4.31)$$

$$\frac{\partial \ln L}{\partial \tilde{\beta}_2} = \frac{1}{\tilde{\sigma}^2} \sum (Y_i - \tilde{\beta}_1 - \tilde{\beta}_2 X_i)(-X_i) \quad \dots (4.32)$$

Setting these equations to zero (i.e., applying the first-order condition for optimization), and re-arranging terms, and denoting by $\tilde{\beta}_1, \tilde{\beta}_2$ and $\tilde{\sigma}^2$ as the 'maximum likelihood estimates (MLEs)', we get:

$$\sum Y_i = n\tilde{\beta}_1 + \tilde{\beta}_2 \sum X_i \quad \dots (4.33)$$

$$\sum Y_i X_i = \tilde{\beta}_1 \sum X_i + \tilde{\beta}_2 \sum X_i^2 \quad \dots (4.34)$$

Equations (4.35) and (4.36) are the same as the normal equations of the OLS method of estimation given at (4.10) and (4.12). Therefore, the maximum likelihood estimators of β_1, β_2 are the same as the OLS estimators.

The maximum likelihood method provides an estimator for the error variance σ^2 also. For taking partial derivative with respect to $\tilde{\sigma}$, let us re-arrange the terms in (4.30) and obtain

$$\begin{aligned} \ln L(\cdot) &= -n \cdot \frac{1}{2} \cdot \ln \sigma^2 - \frac{n}{2} \ln(2\pi) - \frac{1}{2} \sum \frac{(Y_i - \beta_1 - \beta_2 X_i)^2}{\sigma^2} \\ &= -n \cdot \ln \sigma - \frac{n}{2} \ln(2\pi) - \frac{1}{2} \cdot \sigma^{-2} \cdot RSS \\ &[\text{since } \sum (Y_i - \beta_1 - \beta_2 X_i)^2 = \sum \hat{u}_i^2 = RSS] \end{aligned}$$

Thus,

$$\begin{aligned} \frac{\partial \ln L}{\partial \tilde{\sigma}} &= -n \cdot \frac{1}{\tilde{\sigma}} - 0 - \frac{1}{2} \cdot (-2) \cdot \tilde{\sigma}^{-3} \cdot RSS \\ &= -\frac{n}{\tilde{\sigma}} + \frac{RSS}{\tilde{\sigma}^3} \end{aligned}$$

Equating the above to zero yields

$$-\frac{n}{\tilde{\sigma}} + \frac{RSS}{\tilde{\sigma}^3} = 0$$

Or,

$$\frac{RSS}{\tilde{\sigma}^2} = n$$

Or,

$$\tilde{\sigma}^2 = \frac{RSS}{n} \quad \dots (4.35)$$

Remember that the maximum likelihood estimator of σ^2 is biased, but consistent.

4.2.5 Interpretation of OLS Estimators

There could be three types of regression equations:

$$(a) \quad Y_i = \beta_1 + \beta_2 X_i + u_i \quad \dots (4.36)$$

$$(b) \quad \ln Y_i = \beta_1 + \beta_2 X_i + u_i \quad \dots (4.37)$$

$$(c) \quad \ln Y_i = \beta_1 + \beta_2 \ln X_i + u_i \quad \dots (4.38)$$

Equation (4.38) is the usual one, which we considered in (4.1). Equation (4.37) is the semi-log equation where the dependent variable is in log, while the independent variable is in natural numbers. In equation (4.38) we have both the variables in logarithmic terms, and it is called 'double-log' or 'log-linear' model. We have discussed the specification and interpretation of these models in Sub-Section 1.5.2 of Unit 1.

Let us interpret equation (4.36). Suppose we want to find the relationship between wage (w) and years of schooling (s) of workers. For a sample of total

work force in a particular year, let Y = wage measured in INR per hour, and X = education level (expressed in terms of number of years of formal education). Suppose, for a sample of size 600, the regression model obtained is

$$\hat{w} = 0.89 + 0.63s \quad \dots (4.39)$$

where $\hat{w}$ is the predicted value of the wage. The result tells us that ‘when the years of schooling is 0, i.e., the respondent is illiterate, the predicted hourly wage is Rs. 0.89 or 89 paise. If the respondent has 8 years of schooling, the predicted hourly wage is $0.89 + 0.63(8)$ or Rs. 5.93. If the person has 12 years of schooling, his hourly wage will be Rs. 8.45. Thus, when the years of schooling increases by one year, the hourly wage increases by Rs. 0.63 or 63 paise.

Check Your Progress 1

- 1) State the assumptions of the classical regression model.

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- 2) Let the regression model be $Y_i = \beta_1 + \beta_2 X_i + u_i$, where Y is hourly wage rate (measured in rupees per hour) and X is years of schooling. Also, $x_i = (X_i - \bar{X}_i)$ and $y_i = (Y_i - \bar{Y}_i)$.

Based on the following data for a sample of 526 workers, compute the OLS estimators of the intercept β_1 and slope β_2 .

$$n = 526$$

$$\sum_{i=1}^n Y_i = 3101.35$$

$$\sum_{i=1}^n X_i^2 = 87040.0$$

$$\sum_{i=1}^n y_i^2 = 7160.414$$

$$\sum_{i=1}^n X_i = 6608.0$$

$$\sum_{i=1}^n X_i Y_i = 41140.65$$

$$\sum_{i=1}^n x_i^2 = 4025.43$$

$$\sum_{i=1}^n Y_i^2 = 25446.29$$

$$\sum_{i=1}^n x_i y_i = 2179.204$$

$$\sum_{i=1}^n \hat{u}_i^2 = 5980.682$$

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- 3) Interpret the slope coefficient estimated in question (2) above.

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- 4) Specify the likelihood function for the simple regression model.

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4.3 STANDARD ERROR OF THE ESTIMATORS

Recall that the concepts ‘estimator’ and ‘estimate’ are different. While the term estimator relates to the formula or the rule, the term estimate reflects the numerical value. By applying the sample formula, we may obtain different values of $\hat{\beta}_1$ and $\hat{\beta}_2$, as we take different samples. For example, the estimator of $\hat{\beta}_2$ is given by $\hat{\beta}_2 = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$. If we take two different samples, and apply the same estimator (formula), we will get two different estimates (values) of $\hat{\beta}_2$. For hypothesis testing, we need the standard error of the estimators $\hat{\beta}_1$ and $\hat{\beta}_2$.

4.3.1 Standard Error of the OLS Estimators

We need a measure to know the reliability (or precision) of the estimators $\hat{\beta}_1$ and $\hat{\beta}_2$. We need this because the least-square estimators are functions of the sample data and would change with sample. Precision is measured by the standard error (*se*) of the estimators. With the classical assumptions, the variance of the estimators $\hat{\beta}_1$ and $\hat{\beta}_2$ are given below. The standard error of an estimator is the positive square root of its variance.

$$var(\hat{\beta}_2) = \frac{\sigma^2}{\sum x_i^2} \quad \dots (4.40)$$

$$se(\hat{\beta}_2) = \frac{\sigma}{\sqrt{\sum x_i^2}} \quad \dots (4.41)$$

$$var(\hat{\beta}_1) = \frac{\sum x_i^2}{n \sum x_i^2} \sigma^2 \quad \dots (4.42)$$

$$se(\hat{\beta}_1) = \sqrt{\frac{\sum x_i^2}{n \sum x_i^2}} \sigma \quad \dots (4.43)$$